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DETECTING CONSTANT RATES OF CHANGE IN LANGUAGE

AN A PRIORI MONTE CARLO POWER ANALYSIS

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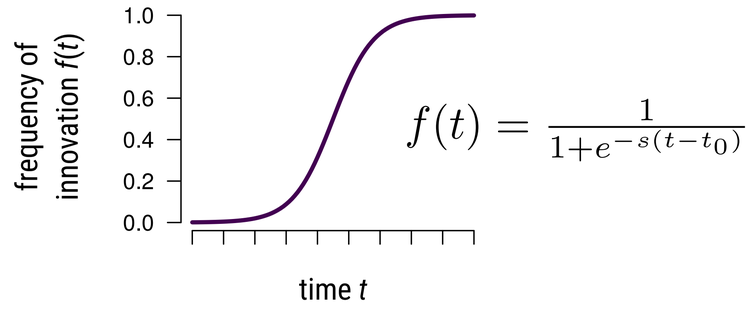
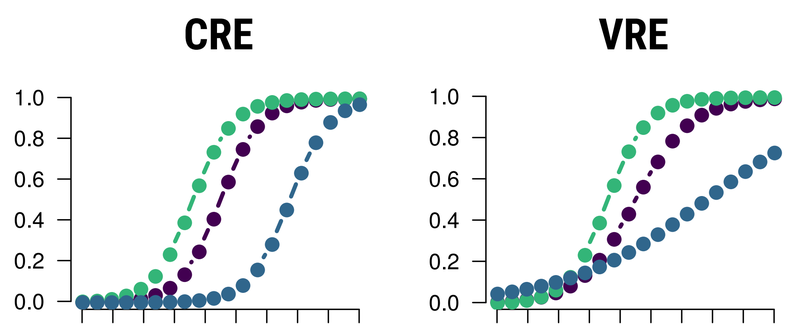
1 / Language change.

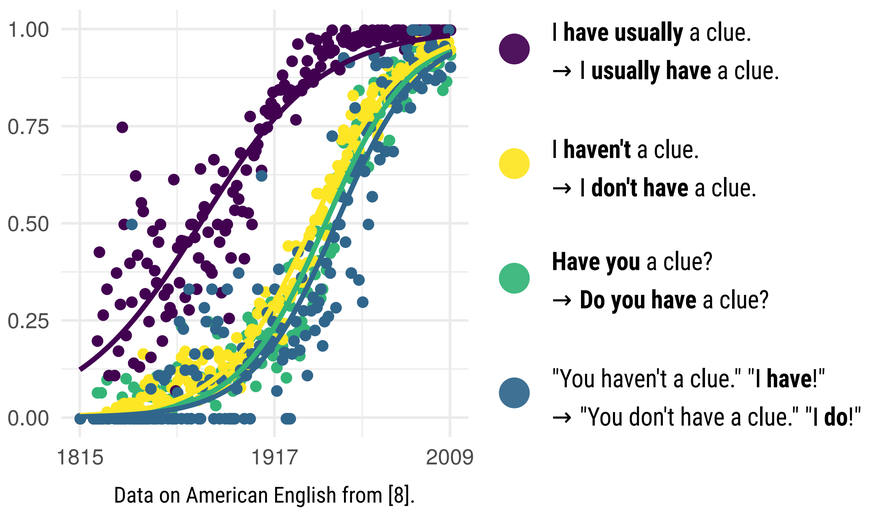
On the population level, language change is sigmoidal and can be modelled using the **logistic function** with two parameters: **slope** s and **intercept** t_0 [1–2]. This is the **Well-Behavedness Postulate (WBP)**.



2 / The Constant Rate Hypothesis.

The **Constant Rate Hypothesis** [2]: Whenever two or more changes are controlled by one and the same mental parameter, the slopes of the logistics are identical. Such cases are **Constant Rate Effects (CREs)**, as opposed to **Variable Rate Effects (VREs)** (*below*). Decision procedure: logistic regression with a time:change interaction term (CRE is the null). Huge literature on this putative **universal of language dynamics** [e.g. 2–7].





Data on American English from [8].

PROBLEM: The type II error rate β of the method is completely unknown [9].

3 / Power analysis: method.

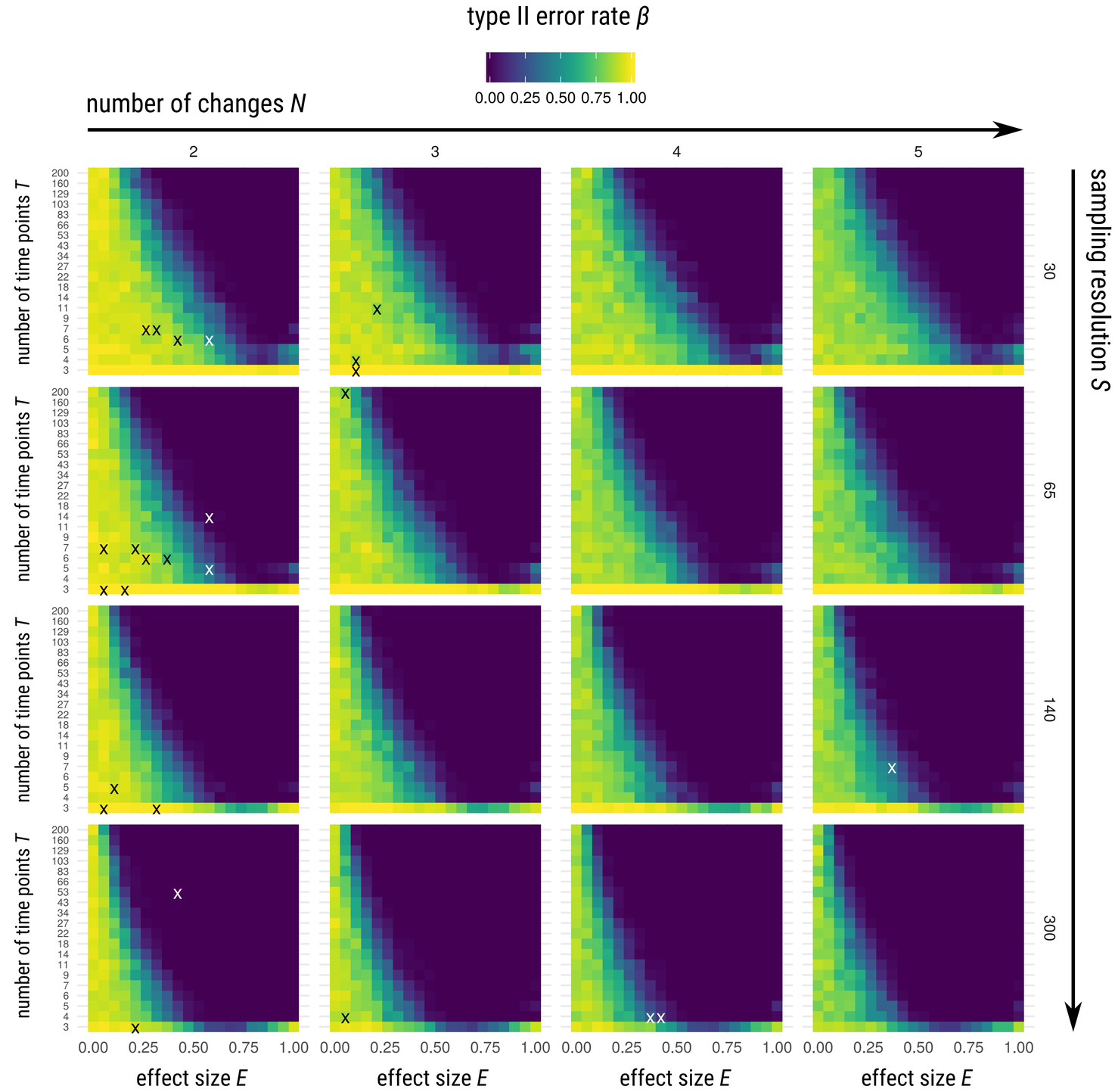
Generate synthetic data from two models:

VRE: both slope s and intercept t_0 are free to vary
CRE: only the intercept t_0 is free to vary.

Cross-fit models using logistic regression. Establish **confusion matrix** over several Monte Carlo repetitions. **Sweep** over the parameters (see *right*):

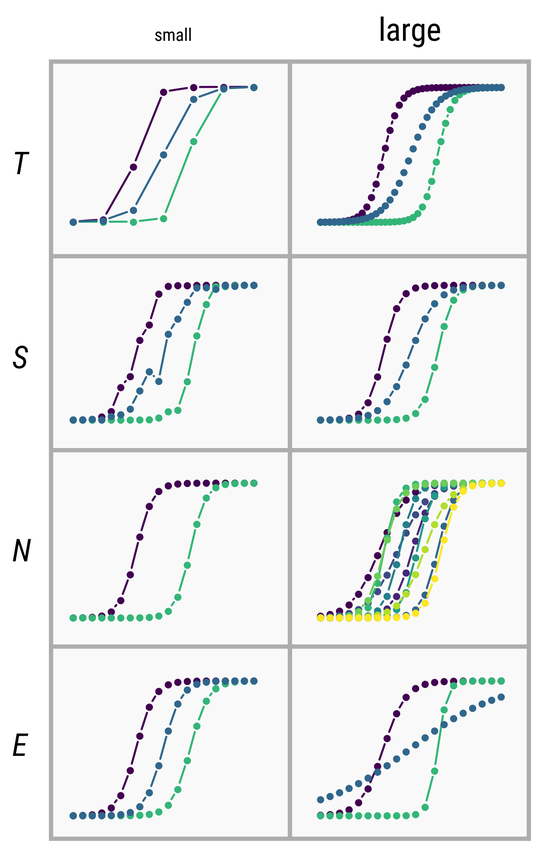
T = **number of time points**
S = **sampling resolution** (how many individual observations make up each time point average)
N = **number of changes**
E = **effect size** (size of VRE), determined as one minus the ratio of the shallowest to the steepest slope:

$$E = 1 - \frac{\min_i(|s_i|)}{\max_i(|s_i|)}$$

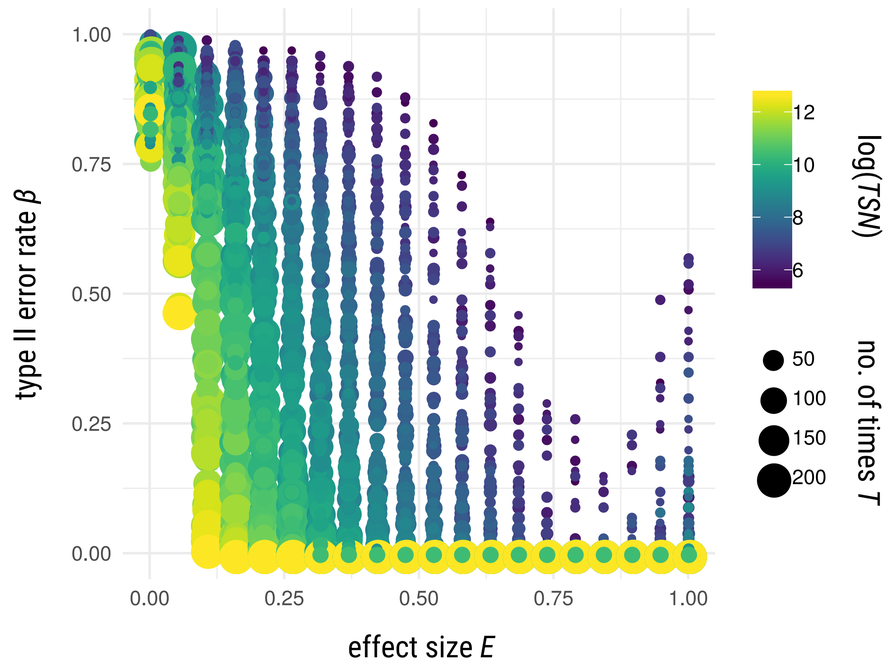


4 / Results.

The type II error rate β (probability of reporting a CRE when in fact a VRE was the generating process) increases with decreasing data resolution parameters (*above*). In fact, β is effectively predicted as a sigmoid function of the size of the VRE (E) and a combined data resolution quantity **log(*TSN*)**, except when the number of times T is very small (*right*).



To estimate the power of existing empirical studies, we compiled a database of 25 putative CREs from the literature. These are shown as crosses overlaid on the parameter sweep (*above*). 22 of the 25 studies attest a type II error rate in excess of 0.2 (power under 0.8).



5 / Conclusions.

(1) CREs can only be reliably diagnosed when the combined **data resolution** is high enough. Many "CREs" reported in the literature to date may not be CREs at all. Studies with low temporal resolution (low T) are particularly vulnerable. **(2)** We need **mechanistic models** of phenomena such as the CRE in order to tease apart true from false positives [10]. **(3)** The WBP may also be false. What would a CRE look like in an **ill-behaved** (non-sigmoid, non-monotonic) change?

References. [1] Blythe, R. A. & Croft, W. 2012. *Language* 88:269–304. [2] Kroch, A. S. 1989. *Language Variation and Change* 1:199–244. [3] Santorini, B. 1993. *Language Variation and Change* 5:257–283. [4] Pintzuk, S. 1995. *Language Variation and Change* 7:229–260. [5] Kallel, A. 2007. *Language Variation and Change* 19:27–49. [6] Durham, M. et al. 2012. *Journal of English Linguistics* 40:316–337. [7] Wallage, P. 2013. *Journal of Historical Syntax* 2:1–25. [8] Zimmermann, R. 2017. PhD thesis, University of Geneva. [9] Paolillo, J. C. 2011. *Language Variation and Change* 23:257–274. [10] Kauhanen, H. & Walkden, G. 2018. *Natural Language & Linguistic Theory* 36:483–521.